

OS Practical No. 5

Aim :

In an operating system three CPU-intensive processes are ready for execution, which require 10ns, 20ns and 30ns and arrival at times 0ns, 2ns and 6ns, respectively. Write a Program to calculate the total number of context switches needed if the operating system implements a shortest job first (preemptive) scheduling algorithm. Also calculate the average time for which the processes have to wait before getting the CPU.

Code :

```
#include <stdio.h>
int main() {
    int n = 3;
    int at[3] = {0, 2, 6};    // Arrival times
    int bt[3] = {10, 20, 30}; // Burst times
    int rt[3];                // Remaining times
    int wt[3] = {0};
    int ct[3];
    int time = 0, completed = 0;
    int min, shortest;
    int prev = -1;
    int context_switch = 0;

    // Copy burst time to remaining time
    for (int i = 0; i < n; i++)
        rt[i] = bt[i];
    while (completed != n)
    {
        min = 9999;
        shortest = -1;

        // Find process with shortest remaining time
        for (int i = 0; i < n; i++) {
            if (at[i] <= time && rt[i] > 0 && rt[i] < min)
            {
                min = rt[i];
                shortest = i;
            }
        }
    }
```

```

    if (shortest == -1)
    {
        time++;
        continue;
    }

    // Count context switches
    if (prev != -1 && prev != shortest)
        context_switch++;
    prev = shortest;

    // Execute process for 1 unit time
    rt[shortest]--;
    time++;

    // If process completes
    if (rt[shortest] == 0) {
        completed++;
        ct[shortest] = time;
        wt[shortest] = ct[shortest] - at[shortest] - bt[shortest];
    }
}

// Display results
float avg_wt = 0;
printf("Process\tAT\tBT\tWT\n");
for (int i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n", i+1, at[i], bt[i], wt[i]);
    avg_wt += wt[i];
}
avg_wt /= n;
printf("\nTotal Context Switches = %d", context_switch);
printf("\nAverage Waiting Time = %.2f ns\n", avg_wt);

return 0;
}

```

devanshu — -zsh — 119x69

```
[devanshu@Devanshus-MacBook-Air ~ % nano gantt.c
[devanshu@Devanshus-MacBook-Air ~ % gcc gantt.c -o gantt_program
[devanshu@Devanshus-MacBook-Air ~ % ./gantt_process
zsh: no such file or directory: ./gantt_process
[devanshu@Devanshus-MacBook-Air ~ % ./gantt_program
Process AT      BT      WT
P1      0       10      0
P2      2       20      8
P3      6       30     24

Total Context Switches = 2
Average Waiting Time = 10.67 ns
devanshu@Devanshus-MacBook-Air ~ %
```